

KAMPHAENG PHET, Thailand

WMO: 483800

Lat: 16.4867N

Lon: 99.5269E

Elev: 268

StdP: 14.55

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	60.9	63.8	53.3	61.0	73.3	56.1	67.8	74.6	7.6	84.8	6.9	83.7	0.1	280	0.305

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	19.7	100.8	78.2	98.8	78.4	97.0	78.1	82.5	92.3	81.8	91.1	81.2	90.1	2.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.1	158.0	86.3	79.6	155.2	85.6	79.1	152.4	84.8	46.7	92.7	45.8	91.4	45.2	90.3	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	5.9	5.0	DB	56.7	103.7	3.2	2.6	54.4	105.6	52.5	107.2	50.7	108.6	48.4	110.5	(4)
(5)			WB	54.8	84.1	3.2	1.5	52.5	85.1	50.7	86.0	48.9	86.8	46.6	87.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.9	77.6	81.2	84.7	87.8	86.4	85.0	83.8	83.6	83.4	82.8	80.9	77.4	(6)	
	DBStd	4.22	3.76	3.39	3.25	3.17	3.51	2.10	2.10	1.66	1.83	2.10	2.86	4.11	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	2	1	0	0	0	0	0	0	0	0	0	0	1	(9)	
	CDD50	12000	856	874	1076	1133	1129	1049	1049	1040	1001	1017	926	850	(10)	
	CDD65	6525	391	454	611	683	664	599	584	575	551	552	476	385	(11)	
	CDH74	73378	3577	4993	7648	9403	8317	6995	6499	6195	5877	5870	4641	3363	(12)	
	CDH80	33301	1472	2462	4107	5423	4234	3003	2566	2298	2231	2304	1896	1305	(13)	
(14) Wind	WSAvg	0.8	0.8	0.8	0.9	1.0	1.1	0.9	0.8	0.7	0.7	0.7	0.7	0.7	(14)	
(15) Precipitation	PrecAvg	51.6	0.2	0.5	1.6	2.1	7.4	6.3	7.1	7.1	10.9	6.8	1.2	0.3	(15)	
	PrecMax	65.4	1.4	2.2	4.8	5.4	15.0	10.0	11.9	13.6	17.8	13.6	4.1	2.2	(16)	
	PrecMin	32.6	0.0	0.0	0.0	0.0	2.0	2.0	2.6	3.4	5.7	0.6	0.0	0.0	(17)	
	PrecStd	8.9	0.4	0.6	1.3	1.5	3.7	2.1	2.6	2.3	3.1	3.2	1.2	0.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.4	98.1	100.8	104.8	103.7	98.1	96.6	94.6	94.6	94.5	94.6	93.5	(19)	
		MCWB	74.1	75.5	75.5	77.2	78.7	79.7	79.3	79.7	80.0	79.8	78.1	75.7	(20)	
	2%	DB	92.2	95.9	98.7	101.7	100.0	95.0	93.7	92.3	92.6	92.5	92.8	91.4	(21)	
		MCWB	73.8	75.3	76.4	78.2	79.4	80.1	79.5	79.9	80.0	79.4	77.8	75.5	(22)	
	5%	DB	90.3	94.1	96.9	99.5	97.3	92.9	91.6	90.7	91.1	91.2	90.9	89.4	(23)	
		MCWB	73.4	75.2	76.2	78.9	79.6	80.1	79.5	79.7	79.9	79.1	77.2	74.6	(24)	
	10%	DB	88.1	92.1	95.1	97.5	94.7	91.0	89.8	89.1	89.4	89.6	89.1	87.2	(25)	
		MCWB	73.0	75.0	76.3	79.0	79.7	80.0	79.4	79.5	79.8	79.0	76.7	73.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.3	81.7	82.0	83.6	83.4	83.1	81.9	81.8	82.4	81.9	81.0	79.8	(27)	
		MCDB	88.6	93.5	93.7	96.3	94.7	92.0	89.9	89.6	89.7	89.6	89.3	87.9	(28)	
	2%	WB	77.0	79.4	80.1	81.9	82.1	82.1	81.0	81.0	81.5	81.0	79.7	77.9	(29)	
		MCDB	86.6	90.7	92.4	95.1	92.3	90.4	88.8	88.6	88.9	88.6	88.3	86.5	(30)	
	5%	WB	75.9	78.1	79.1	80.9	81.2	81.3	80.5	80.5	80.8	80.4	78.8	76.5	(31)	
		MCDB	85.2	89.0	91.3	93.8	91.2	89.2	88.1	87.9	88.2	87.8	87.3	85.2	(32)	
	10%	WB	74.7	77.0	78.2	80.0	80.5	80.6	80.0	79.9	80.1	79.8	78.0	75.4	(33)	
		MCDB	83.8	87.5	90.1	92.5	90.2	88.1	86.9	86.7	87.1	86.7	85.8	83.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	21.3	22.3	21.3	19.7	15.7	13.1	12.4	12.3	12.9	13.8	16.6	19.4	(35)	
		MCDBR	24.0	24.8	24.3	22.4	19.3	15.5	15.1	14.4	15.2	16.0	18.9	20.9	(36)	
	5% WB	MCWBR	8.5	8.4	7.1	5.7	4.6	4.6	4.6	4.5	4.9	5.2	6.2	7.1	(37)	
		MCWBR	19.2	20.9	20.6	19.4	16.2	13.7	12.9	13.0	13.6	14.5	16.6	18.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.481	0.573	0.679	0.616	0.493	0.436	0.428	0.432	0.468	0.550	0.479	0.479	(40)	
		taud	2.085	1.868	1.651	1.838	2.221	2.399	2.421	2.394	2.281	1.999	2.178	2.109	(41)	
	Ebn at Noon		255	237	217	232	258	271	273	275	266	239	254	251	(42)	
		Edh at Noon	49	64	82	68	46	38	37	39	43	56	45	47	(43)	
(44) All-Sky Solar Radiation	RadAvg		1468	1644	1720	1881	1752	1584	1487	1461	1509	1455	1480	1414	(44)	
	RadStd		74	68	113	100	128	125	100	81	85	91	88	58	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.23	N/A	N/A	+193	+194

Nomenclature: See separate page